

Hospital Management System – Requirements

A hospital database system is needed to manage patients, doctors, departments, appointments, medical records, services, and billing:

PATIENT

- Identified by Patient_id.
- Attributes: F_name, L_name, Phone_no, Email, Address, DOB, Age (derived), Blood_group, Gender.
- Can:
 - Schedule multiple appointments with doctors
 - Have multiple medical records
 - Receive multiple bills for services

DOCTOR

- Identified by Doctor_id.
- Attributes: Name, Specialization, Phone_no, Email, License_no, Qualification, Years_of_experience.
- Can:
 - Work in a department
 - Have multiple appointments with patients
 - Create medical records for patients
 - Be supervised by a department head

DEPARTMENT

- Identified by Dept_id.
- Attributes: Dept_name, Location, No_of_doctors, Contact_number.
- A department:
 - Employs multiple doctors
 - One doctor serves as department head (manages the department)
 - Offers multiple services

APPOINTMENT

- Identified by Appointment_id.
- Attributes: Date, Time, Status (Scheduled/Completed/Cancelled), Appointment_type (Consultation/Follow-up/Emergency), Reason.
- Links a patient to a doctor at a specific date and time.
- Can include multiple services provided during the visit.
- Results in a medical record if completed.

SERVICE

- Identified by Service_id.
- Attributes: Service_name, Service_type (Consultation/Lab Test/X-Ray/Surgery/Treatment), Unit_price, Description.
- Offered by departments.
- Multiple services can be provided in a single appointment.
- Each service provided is tracked with quantity (e.g., number of tests).

MEDICAL RECORD

- Identified by Record_id.
- Attributes: Visit_date, Diagnosis, Treatment_plan, Prescribed_medications, Doctor_notes, Follow_up_required.
- Created by a doctor for a patient.
- Associated with an appointment.
- Documents the medical findings and treatment.

BILLING

- Identified by Bill_id.
- Attributes: Bill_date, Total_amount, Payment_status (Paid/Pending/Partial), Payment_method (Cash/Card/Insurance), Due_date.
- Generated for a patient.
- Related to an appointment.
- Calculates total based on:
 - $\text{Services provided during appointment} (\text{Service} \times \text{Quantity} \times \text{Unit_price})$
- Tracks payment information for hospital services.

Key relationship: An appointment can have multiple services provided (many-to-many relationship), and each service provided includes a quantity. The billing then calculates the total cost based on all services rendered during that appointment.